Technical Note

Power Electronics and Control Systems

Six-Pulse Thyristor Bridge (B6C)

with Arm Fuses, RC Snubbers and DC LC Filter

Sizing formula sheet and design path

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Abstract: Formula sheet and design path for a fully controlled six-pulse thyristor bridge (B6C) fed by a transformer, with one fuse in series with each thyristor, an RC snubber across each thyristor and an LC output filter. The sheet collects, without derivation, the relations needed to size the transformer and the no-load voltage, the firing-angle range, the thyristors (voltage, current, thermal, surge), the RC snubbers, the arm fuses and their I^2t coordination, the DC filter and the grid-side quantities, organised in a fourteen-step iterative sizing path. Revision 01 adds the snubber design criteria: reverse-recovery overvoltage from a normalised snap-off transient, voltage budget, capacitance and optimum damping, dv/dt, turn-on discharge, losses and ratings, and the bridge-level protection against the magnetizing energy of the transformer. A worked example (440 V, 1000 A) runs the whole path; its numbers are reproduced by the companion Python script.

Revision history

Rev.	Date	Description	Authors
00	September 2026	First issue.	Davide Bagnara
01	October 2026	RC snubber design criteria (Sec. 5, step S8), bridge-level protection, worked example extended; sizing path renumbered S1–S14; pwr-control template.	Davide Bagnara

Project Repository: repositories-documentation.git

Calculation: Python 3 / NumPy (`b6c_example_calc.py`)

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Scope

Fully controlled six-pulse bridge (B6C, DIN 41761) fed by a transformer (or line reactor), one fuse in series with each thyristor (arm fuse), one RC snubber across each thyristor, and an output LC filter feeding a load with a specified DC voltage ripple. Fundamental frequency f, $\omega = 2\pi f$. Formulas are quoted without derivation; ideal commutation is assumed unless the overlap angle μ appears explicitly. All AC quantities are RMS line-to-line unless noted; the hat denotes a peak value.

Notation

V_d, I_d	DC output voltage, current (mean)	V_1, V_2	primary, secondary line voltage (RMS)
V_{d0}	ideal no-load DC voltage at $\alpha = 0$	$\Delta V_\pm$	grid tolerance $(+/-)$, p.u.
α, μ, γ	firing, overlap, extinction margin angle	u_k, u_r	transformer reactive, resistive p.u. drop
X_k, L_k, R_k	commutation reactance / inductance / resistance (per phase, referred to secondary)	S_T, I_2	transformer rating, secondary current
D_x, D_r, D_t	inductive, resistive, semiconductor voltage drops	V_{T0}, r_T	thyristor threshold voltage, slope resistance
$I_{T(AV)}, I_{T(RMS)}$	thyristor mean, RMS current	I_{TSM}, I^2t_T	surge current (10 ms), thyristor I^2t
I_{Fn}, U_{Fn}	fuse rated current, voltage	I^2t_{pa}, I^2t_{tot}	fuse pre-arcing, total (clearing) I^2t
I''_{k3}, I''_{k2}	three-phase, two-phase initial short-circuit current	$\hat{U}_{arc}$	fuse peak arc voltage
R_s, C_s	arm snubber resistance, capacitance	$Z_0, \omega_{0s}, \zeta_s$	snubber characteristic impedance, natural frequency, damping ratio
Q_{rr}, I_{RM}, S	recovered charge, peak reverse-recovery current, softness factor t_b/t_a	V_{com}, χ	commutation voltage at turn-off, initial-current factor
V_{RM}, k_{rep}	repetitive peak voltage at turn-off, its budget (p.u. of V_{RRM})	L_σ, I_{dis}	snubber loop inductance, turn-on discharge current
i_0, W_m	transformer no-load current (p.u.), magnetizing energy	C_Y	star-equivalent capacitance at the bridge AC terminals
L, C, R	filter inductance, capacitance, load resistance V_d/I_d	f_0, ζ	filter resonance frequency, damping ratio
A_n	peak amplitude of n-th harmonic of the bridge output voltage	r_{pp}	specified peak-to-peak ripple, p.u. of V_d

1 Sizing path

The path is iterative: the transformer no-load voltage depends on the drops, which depend on the transformer; the snubber depends on the commutation inductance, on the firing-angle range and on the voltage class, and may in turn move the voltage class; the fuse coordination depends on the thyristor class, which depends on thermal and fault checks; the filter depends on the worst-case firing angle, which is fixed only after the transformer is chosen. Figure 1 shows the loop structure; the step list below gives the equation references.

S1 Specification: V_d (min/max to be regulated), I_d (nominal, minimum, overload profile I_{ol}, t_{ol}), r_{pp}, $V_1 \pm \Delta V$, f, grid S_{sc}, ambient T_a, cooling, regenerative operation yes/no.

S2 No-load voltage: first guess $d_x = u_k/2$, $d_r \approx 1.05\,u_r$, $D_t \approx 2$–3 V; compute V_{d0} from (2.6), V_2 from (2.1).

S3 Transformer: I_2, S_T (2.7); X_k, R_k, L_k (2.8); recompute D_x, D_r in volts (2.3)–(2.4); back to S2 until V_{d0} converges (2–4 iterations).

S4 Operating points at $V_1(1-\Delta V_-)$, V_1, $V_1(1+\Delta V_+)$: α (3.2), μ (3.1), $\cos\varphi_1$, Q (3.3). Check $\alpha \geq \alpha_{min}$ at the worst point; if α_{max} or Q are unacceptable, change V_2, add taps, or change u_k and return to S2.

S5 Thyristor voltage class (4.1); first transient clamp level (snubber, MOV), confirmed in S8.

S6 Thyristor current and losses (4.3)–(4.4); junction temperature at nominal and overload (4.5)–(4.7); heat-sink requirement (4.6). Fail ⇒ larger device, parallel devices, better cooling.

S7 Thyristor fault and dynamic data from the datasheet: I_{TSM}, I^2t_T (4.8), $(di/dt)_{crit}$, $(dv/dt)_{crit}$, Q_{rr} and I_{RM} versus di/dt at $T_{j,max}$, t_q; check (3.6) with the snubber discharge of S8.

S8 RC snubber (Sec. 5): design point (5.1)–(5.3); C_s from the voltage budget (5.6)–(5.7) and Table 1, R_s (5.8); checks dv/dt (5.9), turn-on discharge (5.10), V_{RM} at the S4 points; losses and ratings (5.11)–(5.13); bridge-level protection (5.14)–(5.15). Fail ⇒ other C_s/R_s, higher voltage class (S5), device with lower Q_{rr} (S7).

S9 Prospective fault currents (6.5)–(6.7) for the fault cases of Sec. 6.2.

S10 Fuse pre-selection: I_{Fn} (6.1), U_{Fn} (6.2), breaking capacity, $\hat{U}_{arc}$ (6.3).



S11 Coordination (6.10)–(6.11): fuse I^2t_{tot} at the actual voltage and fault current against $k_s\, I^2t_T$; non-melting under the overload profile (6.13). Fail ⇒ other fuse, larger thyristor (S7), or larger X_k (S3).
S12 DC filter: $A_6(\alpha_{max})$ (7.1), LC product (7.3), L_{min} for continuous conduction (7.5), f_0 placement (7.7), damping (7.9). Fail ⇒ change the L/C split, add damping, reconsider r_{pp} or α_{max} (S4).
S13 Filter component ratings (7.13)–(7.17); precharge / soft-start.
S14 Verification: grid currents and harmonics (8.1)–(8.2), reactive power compensation, selectivity with the upstream protection, snubber (simulation with a reverse-recovery device model, v_T measured at commissioning), EMC.

2 Transformer and no-load voltage

$$V_{d0} = \frac{3\sqrt{2}}{\pi}\, V_2 = 1.350\, V_2, \qquad V_2 = \frac{\pi}{3\sqrt{2}}\, V_{d0} = 0.7405\, V_{d0} \tag{2.1}$$

$$V_d = V_{d0}\cos\alpha - D_x - D_r - D_t \tag{2.2}$$

$$D_x = \frac{3}{\pi} X_k I_d = \frac{3}{\pi}\omega L_k I_d, \qquad d_x = \frac{D_x}{V_{d0}}\Big|_{I_d = I_{dN}} = \frac{u_k}{2} \tag{2.3}$$

$$D_r = 2R_k I_d, \qquad d_r = \frac{D_r}{V_{d0}}\Big|_{I_d = I_{dN}} = \frac{\pi}{3} u_r \approx 1.047\, u_r \tag{2.4}$$

$$D_t = 2\,(V_{T0} + r_T I_d) \qquad \text{(two devices in series; add fuse and busbar resistive drops if relevant)} \tag{2.5}$$

Required no-load voltage, evaluated at the minimum grid voltage and at the DC current for which the full V_d must be guaranteed (nominal or overload):

$$V_{d0}^{(V_{1,min})} \geq \frac{V_d + D_x + D_r + D_t}{\cos\alpha_{min}}, \qquad V_{d0} = \frac{V_{d0}^{(V_{1,min})}}{1 - \Delta V_-}, \qquad n_{tr} = \frac{V_1}{V_2} \tag{2.6}$$

Transformer ratings (both windings, Dy or Yy, B6 bridge) and commutation impedance:

$$I_2 = \sqrt{\tfrac{2}{3}}\, I_d = 0.816\, I_d, \qquad S_T = \sqrt{3}\, V_2 I_2 = \frac{\pi}{3} V_{d0} I_d = 1.047\, P_{d0} \tag{2.7}$$

$$X_k = u_k \frac{V_2^2}{S_T} + X_L + X_{grid}, \qquad R_k = u_r \frac{V_2^2}{S_T}, \qquad L_k = \frac{X_k}{\omega}, \qquad X_{grid} = \frac{V_2^2}{S_{sc}} \text{ (referred to secondary)} \tag{2.8}$$

X_k is the total commutation reactance per phase: transformer leakage, line reactor and grid impedance referred to the bridge terminals. The transformer thermal rating must also cover the overload profile and the harmonic content (K-factor or manufacturer derating).

3 Commutation, control angle, power factor

$$\cos\alpha - \cos(\alpha + \mu) = \frac{\sqrt{2}\, X_k I_d}{V_2} = \frac{2\omega L_k I_d}{\sqrt{2}\, V_2} \tag{3.1}$$

$$\alpha(V_1) = \arccos \frac{V_d + D_x + D_r + D_t}{V_{d0}(V_1)}, \qquad \alpha_{max} = \alpha(V_{1,max},\, I_{d,min}) \text{ (largest ripple)} \tag{3.2}$$

$$\cos\varphi_1 \approx \frac{\cos\alpha + \cos(\alpha+\mu)}{2}, \qquad \lambda = \frac{P}{S} \approx \frac{3}{\pi}\cos\varphi_1 = 0.955\cos\varphi_1, \quad Q_1 = P\tan\varphi_1, \quad P = V_d I_d \tag{3.3}$$

Limits on α:

$$\alpha_{min} \approx 5\text{–}10^\circ \text{ (firing reliability, control margin at } V_{1,min}) \tag{3.4}$$

$$\alpha + \mu + \gamma \leq 180^\circ, \qquad \gamma \geq \omega t_q + \Delta\gamma \text{ (inverter mode only)} \tag{3.5}$$

With a unipolar filter capacitor on the DC side, inverter operation ($V_d < 0$) is not possible; α is then limited to the rectifier range and γ is not a constraint.

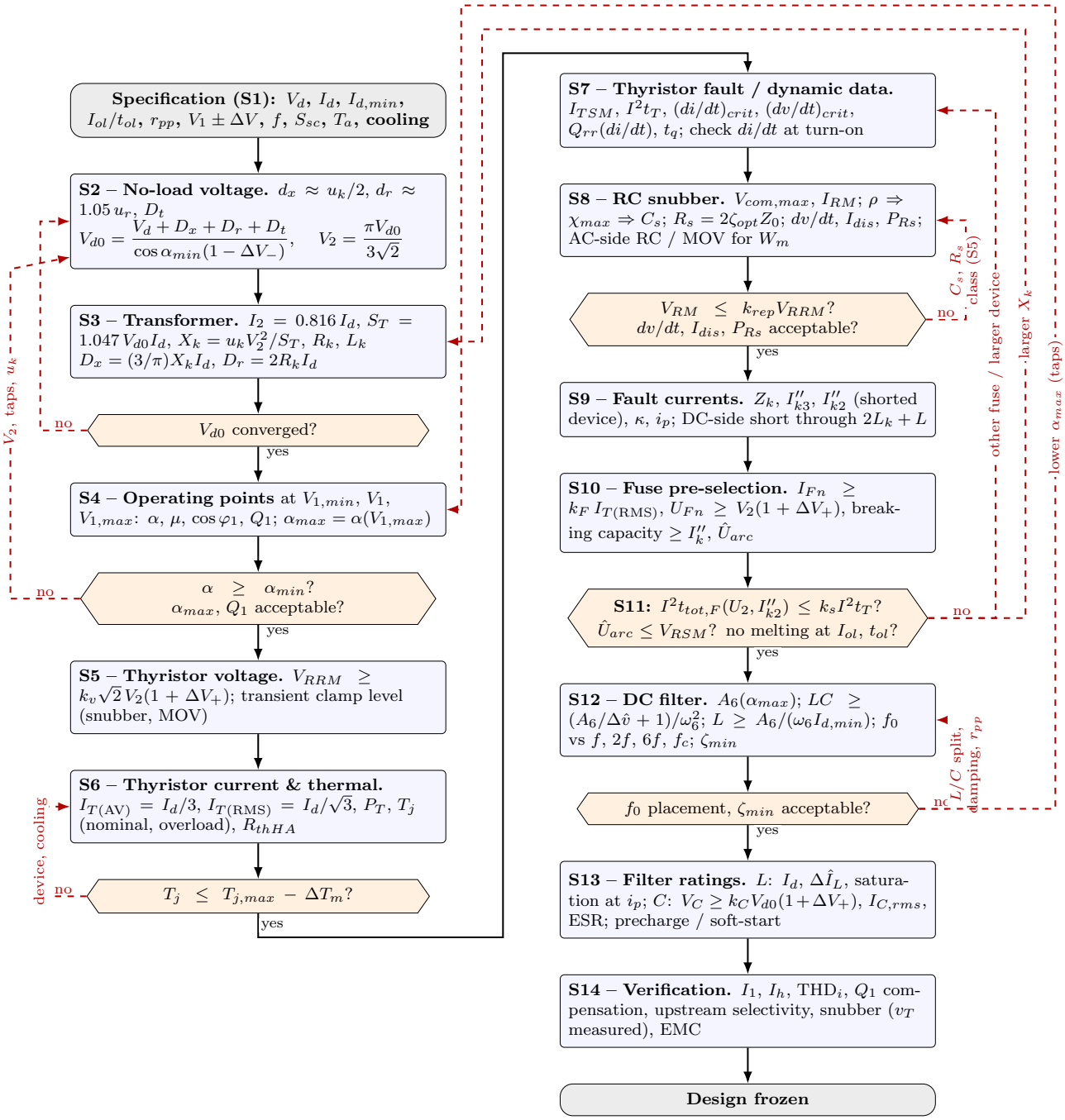


Figure 1: Sizing path. Solid: forward steps; dashed red: iteration loops. Left column S1–S6, right column S7–S14.

Turn-on current slope of the incoming thyristor (natural commutation, snubber discharge (5.10) added separately):

$$\left(\frac{di}{dt}\right)_{on} = \frac{\sqrt{2}\,V_2 \sin\alpha}{2L_k} + \left(\frac{di}{dt}\right)_{snubber} \le \left(\frac{di}{dt}\right)_{crit} \tag{3.6}$$

4 Thyristor selection

4.1 Voltage

$$\hat{V}_{2,max} = \sqrt{2}\,V_2(1+\Delta V_+), \qquad V_{RRM} = V_{DRM} \ge k_v\,\hat{V}_{2,max}, \qquad k_v = 2.0\ldots2.5 \tag{4.1}$$

$$V_{clamp} \le V_{RRM} \text{ (MOV / snubber transient level)}, \qquad V_{RSM} \ge \hat{U}_{arc}(U_2) \text{ (fuse arc voltage, Sec. 6)} \tag{4.2}$$

$k_v \approx 2.0$ is acceptable when transients are clamped and verified; $k_v \approx 2.5$ otherwise. Standard classes: 1200, 1400, 1600, 1800, 2200, 2800, 4500, 6500 V. The class also fixes the room left to the snubber: the repetitive peak at turn-off is $F\,\hat{V}_{2,max}\sin(\alpha+\mu)_{max}$ with $F \approx 1.2\ldots1.5$ (Sec. 5), so that with $k_v = 2$ and a bridge that can run at $\alpha+\mu \approx 90°$ the budget (5.6) cannot be met with a practical snubber.

4.2 Current and losses

$$I_{T(\mathrm{AV})} = \frac{I_d}{3}, \qquad I_{T(\mathrm{RMS})} = \frac{I_d}{\sqrt{3}}, \qquad \text{form factor} = \sqrt{3}\ (120° \text{ rectangular}) \tag{4.3}$$

$$P_T = V_{T0}I_{T(\mathrm{AV})} + r_T I^2_{T(\mathrm{RMS})} = \frac{V_{T0}I_d}{3} + \frac{r_T I_d^2}{3}\ (+P_{sw}+P_G,\ \text{negligible at } 50\,\mathrm{Hz}) \tag{4.4}$$

The datasheet $I_{T(\mathrm{AV})}$ is defined for a 180° half-sine at a given case temperature; do not compare it directly with $I_d/3$, use (4.4) with the thermal chain instead.

4.3 Thermal

$$T_j = T_a + P_T\,(R_{thJC} + R_{thCH} + R_{thHA}) \qquad \text{or} \qquad T_j = T_{hs} + P_T\,(R_{thJC} + R_{thCH}) \tag{4.5}$$

$$R_{thHA} \le \frac{T_{j,max} - \Delta T_m - T_a}{P_T} - R_{thJC} - R_{thCH}, \qquad \Delta T_m = 15\text{–}25\,\mathrm{K} \tag{4.6}$$

$$T_j(t_{ol}) = T_{hs} + P_T R_{thJH} + (P_{ol} - P_T)\,Z_{thJH}(t_{ol}), \qquad P_{ol} = \frac{V_{T0}I_{ol}}{3} + \frac{r_T I_{ol}^2}{3} \tag{4.7}$$

T_{hs} constant if $t_{ol} \ll \tau_{hs}$; otherwise add $(P_{ol}-P_T)\,Z_{thHA}(t_{ol})$. For cyclic loads check ΔT_j against the power-cycling capability. With N_p devices in parallel use I_d/N_p with a current-sharing factor (typ. 1.1–1.2).

4.4 Surge, I^2t, t_q

$$I^2t_T = \left.\frac{I^2_{TSM}\,t_p}{2}\right|_{t_p=10\,\mathrm{ms}} \text{ (half-sine)}, \qquad I^2t_T(t) \approx I^2t_T\sqrt{\frac{t}{10\,\mathrm{ms}}} \quad (t > 10\,\mathrm{ms},\ I^2\sqrt{t}\ \text{rule}) \tag{4.8}$$

$$t_q \le \frac{\gamma_{min}}{\omega}\ \text{(inverter mode)} \tag{4.9}$$

The dynamic data that size the snubber, $(dv/dt)_{crit}$, $(di/dt)_{crit}$, Q_{rr} and I_{RM} versus di/dt at $T_{j,max}$, are used in Sec. 5.

5 RC snubbers

5.1 Function and connection

One series R_s–C_s branch across each thyristor (arm snubber), mounted at the device terminals, inside the arm fuse. It must

(i) limit the repetitive overvoltage produced when the reverse-recovery current of the outgoing thyristor snaps off in the commutation loop $2L_k$;
(ii) keep the rate of rise of forward voltage on the blocking thyristors (commutation notches of the other arms, fast grid transients) below $(dv/dt)_{crit}$;
(iii) do so without overloading the thyristor at turn-on (capacitor discharge) or the resistor (losses).

Non-repetitive overvoltages (switching of the transformer, grid surges) carry far more energy than the arm snubbers can take and are handled by the bridge-level protection of Sec. 5.4. Inside the fuse, the turn-on discharge flows through the thyristor only, and an arm whose fuse has cleared is isolated; a snubber across fuse and thyristor would leave an R_s–C_s path ($\approx \omega C_s V_2$) across the open arm. The arm snubbers do not limit the fuse arc voltage: (6.3) remains the check. The inductance L_σ of the loop thyristor–C_s–R_s is not covered by the snubber ($L_\sigma\, di/dt$ appears directly on the device): keep it to tens of nH.

5.2 Recovery transient

$$V_{com} = \sqrt{2}\, V_2 \sin(\alpha+\mu), \qquad \left(\frac{di}{dt}\right)_{off} = \frac{V_{com}}{2L_k} \quad \text{(outgoing thyristor at the end of the overlap)} \tag{5.1}$$

$$I_{RM} \approx \sqrt{\frac{2\,Q_{rr}\,(di/dt)_{off}}{1+S}}, \qquad Q_{rr} \approx Q_{rr,ref}\sqrt{\frac{(di/dt)_{off}}{(di/dt)_{ref}}} \quad \text{(datasheet at } T_{j,max}\text{; scaling only if one point is given)} \tag{5.2}$$

$$E_L = \tfrac{1}{2}\,(2L_k)\,I_{RM}^2 = \frac{Q_{rr}V_{com}}{1+S} \quad \text{(energy released at snap-off, per commutation)} \tag{5.3}$$

V_{com}, $(di/dt)_{off}$ and I_{RM} all grow with $\alpha+\mu$. If the bridge can run at $\alpha \approx 90°$ (soft start from $\alpha \approx 90°$, current limit, transients), the design point is $\alpha+\mu = 90°$: $V_{com,max} = \hat{V}_{2,max}$, with I_d of that condition (nominal or current limit) and Q_{rr} at $T_{j,max}$; otherwise $(\alpha+\mu)_{max}$ from S4. The other operating points are checked with the chosen R_s, C_s.

The method follows [7, 3, 8] and the diode-bridge note [9], extended to the firing angle. At snap-off the recovery current is diverted into the snubber (ideal snap-off: conservative for soft-recovery devices). Equivalent circuit: V_{com}, $2L_k$, R_s, C_s in series, $i(0) = I_{RM}$, $v_C(0) = 0$:

$$Z_0 = \sqrt{\frac{2L_k}{C_s}}, \qquad \omega_{0s} = \frac{1}{\sqrt{2L_kC_s}}, \qquad \zeta_s = \frac{R_s}{2Z_0}, \qquad \chi = \frac{I_{RM}Z_0}{V_{com}} \tag{5.4}$$

$$v_T(0^+) = R_s I_{RM}, \qquad V_{RM} = F(\zeta_s,\chi)\,V_{com}, \qquad F\big|_{R_s=0} = 1+\sqrt{1+\chi^2} \quad \text{(undamped bound)} \tag{5.5}$$

$F(\zeta_s,\chi)$ is given in Table 1 and Fig. 2 (normalisation of McMurray [7]). For $\chi \gtrsim 0.5$ there is an optimum damping $\zeta_{opt}(\chi)$: below it the $2L_k$–C_s oscillation dominates the peak, above it the initial step R_sI_{RM}. For $\chi < 0.5$ (large V_{com}, small recovery charge) the peak falls monotonically with ζ_s, and the damping is limited by (5.9) and (5.10) instead. The table reproduces the reference points of [3] ($\chi = 1$: $F_{min} = 1.5$ at $R_s \approx 1.3\,Z_0$).

Table 1: Normalised repetitive peak $F = V_{RM}/V_{com}$ of the snap-off transient (5.5), and optimum damping. Computed with `b6c_example_calc.py`.

$\zeta_s \setminus \chi$	0	0.25	0.5	0.75	1.0	1.25	1.5	2.0	2.5	3.0
0.3	1.45	1.47	1.52	1.60	1.72	1.86	2.01	2.35	2.72	3.09
0.4	1.36	1.38	1.42	1.50	1.61	1.75	1.91	2.27	2.65	3.05
0.5	1.30	1.31	1.35	1.43	1.55	1.70	1.88	2.28	2.72	3.18
0.6	1.25	1.26	1.30	1.38	1.51	1.69	1.91	2.43	3.00	3.60
0.7	1.21	1.22	1.26	1.35	1.51	1.77	2.10	2.80	3.50	4.20
0.8	1.18	1.19	1.22	1.33	1.60	2.00	2.40	3.20	4.00	4.80
1.0	1.14	1.14	1.18	1.50	2.00	2.50	3.00	4.00	5.00	6.00
ζ_{opt}	–	–	1.09	0.79	0.64	0.56	0.51	0.44	0.40	0.38
F_{min}	–	–	1.18	1.33	1.51	1.69	1.88	2.26	2.65	3.05

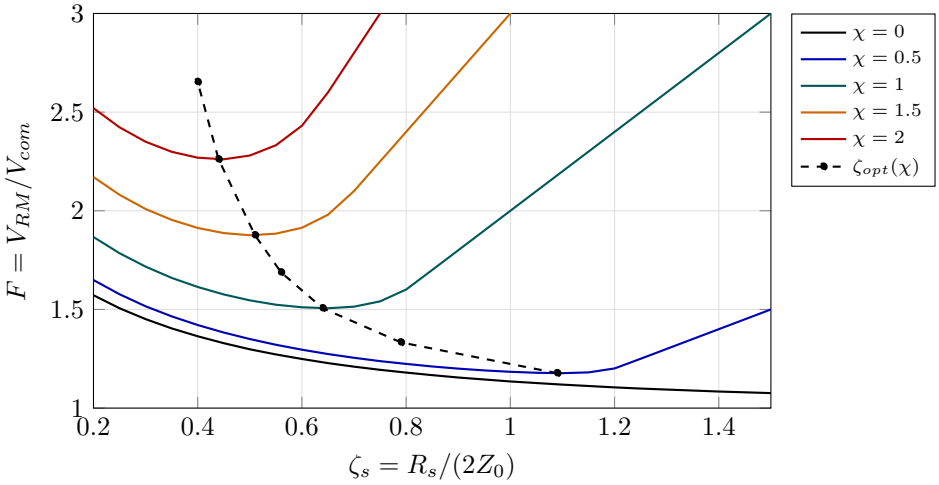


Figure 2: Normalised peak voltage at snap-off versus damping, for several initial-current factors $\chi = I_{RM} Z_0 / V_{com}$. On the right of each minimum the curve becomes the straight line $2\zeta_s \chi = R_s I_{RM}/V_{com}$ (initial step).

5.3 Sizing criteria

C1 Voltage budget. The repetitive peak keeps a margin for the non-repetitive transients clamped at bridge level, for the spread of Q_{rr} and for L_σ:

$$V_{RM} \le V_{RM,adm} = k_{rep} V_{RRM}, \quad k_{rep} = 0.5 \ldots 0.6, \qquad \rho = \frac{V_{RM,adm}}{V_{com,max}} = \frac{k_{rep}\, k_v}{\sin(\alpha+\mu)_{max}} \tag{5.6}$$

C2 Capacitance. Largest admissible χ from $F_{min}(\chi_{max}) = \rho$ (Table 1), then

$$C_s \ge 2L_k \left(\frac{I_{RM}}{\chi_{max} V_{com,max}} \right)^2, \qquad C_s \ge 2L_k \left(\frac{I_{RM}}{\Delta V_C} \right)^2, \ \Delta V_C = V_{RM,adm} - V_{com} \text{ (energy bound)} \tag{5.7}$$

The energy bound needs no table and is always on the safe side, by a factor of about 3 to 5 on C_s for $\rho = 1.3 \ldots 2$; it is the natural choice for diode bridges and small α, where V_{com} is small and χ large. As $V_{com} \to \hat{V}_{2,max}$ the step V_{com} alone overshoots ($F \ge 1.2$ for practical damping), and below $\rho \approx 1.3$ no practical snubber meets the budget: raise the class (S5).

C3 Resistance. Optimum damping for the chosen C_s:

$$R_s = 2\,\zeta_{opt}(\chi)\, Z_0 \ (\zeta_{opt} \approx 0.5 \ldots 0.8, \text{ i.e. } R_s \approx (1 \ldots 1.6)\, Z_0, \text{ for } \chi = 0.75 \ldots 1.5) \tag{5.8}$$

A larger R_s (as in tables for small bridges) builds the overvoltage itself through the step $R_s I_{RM}$.

C4 dv/dt**.** Forward voltage steps ΔV applied through $2L_k$ to a blocking thyristor (initial current zero):

$$\left(\frac{dv}{dt}\right)_{max} = g(\zeta_s)\, \omega_{0s} \Delta V \overset{\zeta_s \ge 0.5}{=} \frac{R_s \Delta V}{2L_k} \le \frac{(dv/dt)_{crit}}{k_{dv}}, \quad k_{dv} \approx 2 \ \Rightarrow \ R_s \le \frac{2L_k\,(dv/dt)_{crit}}{k_{dv}\,\hat{V}_{2,max}} \tag{5.9}$$

$g = 2\zeta_s$ for $\zeta_s \ge 0.5$ (maximum at $t = 0^+$), $g \approx 0.8 \ldots 0.9$ for $\zeta_s = 0.3 \ldots 0.4$. $\Delta V = \hat{V}_{2,max}$ is conservative: the notches caused by the commutations of the other arms are about $\frac{1}{2}\hat{V}_2 \sin\alpha$ per step. The step $R_s I_{RM}$ at snap-off is in the reverse direction and is covered by C1; in rectifier operation the forward voltage returns on the outgoing device at grid rate.

C5 Turn-on discharge. At firing C_s, charged to $V_T(\alpha) \approx \hat{V}_2 \sin\alpha$, discharges through R_s and the thyristor:

$$I_{dis} = \frac{V_T(\alpha)}{R_s} \le \frac{\hat{V}_{2,max}}{R_s}, \qquad \left(\frac{di}{dt}\right)_{dis} \approx \frac{V_T}{L_\sigma} \text{ for } t \lesssim \frac{L_\sigma}{R_s}, \qquad E_{dis} = \tfrac{1}{2} C_s V_T^2 \tag{5.10}$$

I_{dis} adds to (3.6); it must stay small compared with the on-state current at which $(di/dt)_{crit}$ is specified and within the manufacturer's recommendation, which gives a lower limit $R_s \ge \hat{V}_{2,max}/I_{dis,adm}$. An I_{dis}

above the latching current helps firing at light load with a large DC inductance. R_s must lie in the window set by (5.10) and (5.9).

C6 Losses. Per cycle each snubber sees the turn-on discharge, the turn-off step, and two half-steps for each of the three commutations of the other arms that move the voltage of the blocking device; a step ΔV dissipates $\frac{1}{2}C_s\Delta V^2$ in R_s whatever R_s:

$$P_{Rs} \approx f\left[\tfrac{7}{8}\,C_s\hat{V}_2^2\left(\sin^2\alpha+\sin^2(\alpha+\mu)\right)+E_L\right] \le f\left(1.75\,C_s\hat{V}_2^2+E_L\right) \tag{5.11}$$

The charging current at grid frequency, $\omega C_s\hat{V}_2$, adds $R_s(\omega C_s\hat{V}_2)^2/2$ (negligible). Rate R_s for the steady-state worst case (α_{max}) and check its thermal transient at $\alpha \approx 90°$ (soft start, current limit). A switched simulation of the bridge with ideal thyristors and the data of the worked example (`b6c_snubber_sim.py`) gives P_{Rs} within 6 % of (5.11) without E_L for $\alpha = 10\ldots80°$, a forward dv/dt below 40 % of the bound (5.9), and at $\alpha = 80°$ a reverse peak of $1.24\,\hat{V}_2$ without recovery, against $1.22\,\hat{V}_2$ from Table 1 ($\chi = 0$).

C7 Ratings.

$$C_s:\quad V_C \ge V_{RM}\ \text{(repetitive peak; AC component} \approx \hat{V}_{2,max}),\qquad \hat{\imath}_C = \max(I_{RM},\, I_{dis})$$
$$I_{Cs,rms} = I_{Rs,rms} = \sqrt{P_{Rs}/R_s} \tag{5.12}$$

$$R_s:\quad P_{Rs}\ \text{at}\ \alpha_{max},\qquad E_{pulse} = \max\!\left(\tfrac{1}{2}C_s\hat{V}_{2,max}^2,\ E_L+\tfrac{1}{2}C_sV_{com}^2\right),\qquad \text{non-inductive} \tag{5.13}$$

Polypropylene film, self-healing, snubber-rated (pulse $dv/dt \ge \hat{\imath}_C/C_s$), low ESL. With N_p devices in parallel and one snubber per device, size each one with $2L_k \to N_p\,2L_k$ and $(di/dt)_{off} \to (di/dt)_{off}/N_p$ (equal sharing).

If C1 cannot be met with acceptable C_s and losses: higher voltage class (S5), device with lower Q_{rr} or softer recovery (S7). The values belong to the pair snubber–commutation inductance: a change of transformer, line reactor or grid short-circuit power changes L_k and requires a new check.

5.4 Bridge-level protection

Switching off the transformer at no load (primary breaker) releases the magnetizing energy into the capacitance on the secondary side. Upper bound (all the energy into the capacitors, worst distribution among the phases):

$$W_m \approx \frac{i_0\,S_T}{2\omega}\ \text{(three-phase, linear magnetizing inductance)},\qquad \hat{V}_{max}^2 = \hat{V}_{2,max}^2 + \frac{4W_m}{C_Y} \tag{5.14}$$

$$C_Y \ge \frac{2\,i_0\,S_T}{\omega\,(\hat{V}_{max}^2-\hat{V}_{2,max}^2)}\ \text{(star-equivalent capacitance per phase at the bridge AC terminals, } \hat{V}\ \text{line-to-line)} \tag{5.15}$$

The six arm snubbers give $C_Y \approx 2C_s$ (with p and n at the neutral potential), usually far too little. Either an AC-side RC network (star or delta, with a series damping resistor) sized with (5.15), or MOVs with energy rating $\ge W_m$ (with margin for the number of operations) and clamping voltage below V_{RSM}. $\hat{V}_{max} \approx (0.7\ldots0.8)\,V_{RRM}$ is a reasonable target for an event that recurs at every switch-off. MOVs or the AC-side RC also clamp grid surges; an earthed screen between the windings limits the transients transferred capacitively at energization.

6 Arm fuses

6.1 Ratings

$$I_{Fn} \ge k_F\,I_{T(\mathrm{RMS})} = k_F\,\frac{I_d}{\sqrt{3}},\qquad k_F = k_T\,k_{cool}\,k_{cyc} \approx 1.2\ldots1.5 \tag{6.1}$$

$$U_{Fn} \ge V_2(1+\Delta V_+)\ \text{(AC rating; the arm sees the line voltage)},\qquad I_{cn} \ge I''_{k,max}\ \text{(breaking capacity)} \tag{6.2}$$

$$\hat{U}_{arc}(U_2) \le V_{RSM}\ \text{(conservative; better} \le V_{RRM}) \tag{6.3}$$

$$P_F \approx P_{Fn}\left(\frac{I_{T(\mathrm{RMS})}}{I_{Fn}}\right)^m,\ m \approx 1.6\ldots2\ \text{(fuse losses, add to cooling; fuse temperature reduces } I^2t_{pa}) \tag{6.4}$$

k_T: ambient / adjacent-component temperature; k_{cool}: forced-air or liquid cooling of the fuse (manufacturer data); k_{cyc}: cyclic-load fatigue derating (operation at 0.6–0.8 of I_{Fn} for strongly cyclic duty). In a symmetric bridge the arc overvoltage divides over the loop inductances, so healthy devices see only part of $\hat{U}_{arc}$; (6.3) is the usual conservative rule unless a fault-loop simulation is available.

6.2 Fault cases and prospective currents

$$Z_k = \sqrt{R_k^2 + X_k^2}, \qquad I''_{k3} = \frac{V_2(1+\Delta V_+)}{\sqrt{3}\,Z_k} \approx \frac{I_{2N}}{u_k}(1+\Delta V_+) \text{ (DC-side short through the bridge)} \tag{6.5}$$

$$I''_{k2} = \frac{V_2(1+\Delta V_+)}{2Z_k} = \frac{\sqrt{3}}{2} I''_{k3} \text{ (shorted device / misfiring: line-to-line via two arms)} \tag{6.6}$$

$$\kappa = 1.02 + 0.98\, e^{-3R_k/X_k}, \qquad i_p = \kappa\sqrt{2}\, I''_k \text{ (peak asymmetric current, no fuse limiting)} \tag{6.7}$$

$$\left(\frac{di}{dt}\right)_{fault} \approx \frac{\sqrt{2}\,V_2(1+\Delta V_+)}{2L_k} \text{ (device fault)}, \quad \left(\frac{di}{dt}\right)_{fault,DC} \approx \frac{\sqrt{2}\,V_2(1+\Delta V_+)}{2L_k + L} \text{ (DC-side short, first instants)} \tag{6.8}$$

Cases: (a) thyristor failure or misfiring → two-phase short through the failed arm and the next fired arm, limited by $2Z_k$ only (worst case for the fuse); (b) DC-side short circuit → three-phase fed, additionally limited by L, shared by the conducting arms; (c) capacitor discharge into a load short does not flow through the arm fuses but stresses L and the load wiring; (d) commutation failure in inverter mode (if applicable).

6.3 I^2t coordination

$$I^2t_{tot,F}(U_2) = I^2t_{pa} + I^2t_{arc}(U_2) = K_U\left(\tfrac{U_2}{U_{Fn}}\right)\cdot I^2t_{tot,F}(U_{Fn}), \qquad K_U < 1 \text{ (manufacturer curve, incl. } \cos\varphi_k) \tag{6.9}$$

$$I^2t_{tot,F}(U_2, I''_{k2}) \le k_s\, I^2t_T, \qquad k_s = 0.7\ldots0.8 \text{ (tolerances, } T_j \text{ above the datasheet condition)} \tag{6.10}$$

$$\hat{\imath}_c(I''_{k2}) \le I_{TSM} \text{ (cut-off current from the current-limiting curve)} \tag{6.11}$$

$$t_m(I) \approx \frac{I^2t_{pa}}{I^2} \text{ } (t_m \lesssim 10\,\text{ms, adiabatic}) \tag{6.12}$$

$$t_m\left(\frac{I_{ol}}{\sqrt{3}}\right) \ge (3\ldots5)\, t_{ol} \text{ (no melting under overload; time–current curve)} \tag{6.13}$$

For fault durations beyond one half-cycle compare the fuse time–current characteristic with the $I^2\sqrt{t}$ curve of the thyristor (4.8), not with a constant I^2t_T. Upstream selectivity: the arm fuse must clear an internal fault before the upstream breaker or transformer fuse operates; the upstream device must clear the faults the arm fuse cannot see (transformer, busbars).

7 Output LC filter

7.1 Harmonic content of the bridge output voltage

Harmonics of order $n = 6k$ of the fundamental, ideal commutation ($\mu = 0$, conservative), peak amplitude:

$$A_n(\alpha) = \frac{2V_{d0}}{n^2-1}\sqrt{1 + (n^2-1)\sin^2\alpha}, \qquad \omega_n = n\omega, \qquad A_6(0) = 0.0571\,V_{d0}, \quad A_6(90^\circ) = 0.343\,V_{d0} \tag{7.1}$$

Evaluate at α_{max} (maximum grid voltage, minimum load) with $V_{d0}(V_{1,max})$; the sixth harmonic dominates. Unfiltered peak-to-peak ripple $\approx 2A_6$. Non-characteristic components at f and $2f$ appear with grid unbalance and firing asymmetry and are *not* attenuated by a filter tuned below $6f$; keep f_0 away from them.

7.2 Attenuation and LC product

Load approximated by $R = V_d/I_d$ in parallel with C:

$$H(j\omega) = \frac{\hat{v}_C}{\hat{v}_d} = \frac{1}{1 - \omega^2 LC + j\omega L/R}, \qquad |H(\omega_n)| \approx \frac{1}{\omega_n^2 LC - 1} \ (\omega_n \gg \omega_0) \tag{7.2}$$

$$\Delta\hat{v}_{target} = \frac{r_{pp} V_d}{2}, \qquad LC \geq \frac{1}{\omega_6^2}\left(\frac{A_6(\alpha_{max})}{\Delta\hat{v}_{target}} + 1\right) \tag{7.3}$$

$$\Delta v_{pp} \approx 2\sum_k A_{6k}\,|H(\omega_{6k})| \ \text{(check the 12th; usually} < 10\,\% \text{ of the 6th contribution)} \tag{7.4}$$

7.3 Inductance: continuous conduction and ripple current

$$\Delta\hat{I}_L = \frac{A_6}{|\omega_6 L - 1/(\omega_6 C)|} \approx \frac{A_6}{\omega_6 L}, \qquad L \geq L_{min} = \frac{A_6(\alpha)}{\omega_6\, I_{d,min}} \ (\text{continuous conduction at } I_{d,min}) \tag{7.5}$$

$$L_{min}\big|_{\alpha=0,\ \text{resistive load}} = \frac{R_{max}}{105\,\omega} \ (\text{classic critical inductance}), \qquad L \approx (1.2 \ldots 1.5)\, L_{min} \ (\text{tolerance, saturation}) \tag{7.6}$$

7.4 Resonance placement and damping

$$f_0 = \frac{1}{2\pi\sqrt{LC}}, \qquad f_0 \leq \frac{6f}{3} \ (\text{attenuation} \geq 8 \text{ at } 6f) \tag{7.7}$$

$$f_0 \notin (0.8 \ldots 1.25) \times \{f,\, 2f\} \ (\text{non-characteristic components}), \qquad f_0 \geq (3 \ldots 5)\, f_c \ (\text{current-loop bandwidth}) \tag{7.8}$$

$$\zeta = \frac{1}{2R}\sqrt{\frac{L}{C}} \ (\text{load damping, worst at light load } R_{max}), \qquad \zeta_{min} \geq 0.3 \ldots 0.5, \qquad |H|_{max} \approx \frac{1}{2\zeta\sqrt{1-\zeta^2}} \tag{7.9}$$

$$R_{d,opt} = \sqrt{\frac{L}{C}}\sqrt{\frac{(2+n)(4+3n)}{2n^2(4+n)}}, \qquad |H|_{max} = \sqrt{\frac{2(2+n)}{n}}, \qquad n = \frac{C_d}{C} \ (R_d\text{–}C_d \text{ branch parallel to } C) \tag{7.10}$$

$$\zeta_p = \frac{1}{2R_p}\sqrt{\frac{L}{C}} \ (R_p \text{ parallel to } L\text{; zero at } R_p/L\text{, attenuation} \to 1/(\omega R_p C) \text{ above it}) \tag{7.11}$$

$$|H(j\omega)|,\ |H(j2\omega)| \ \text{evaluated with } \zeta_{min} \ (f,\, 2f \text{ components from unbalance and firing asymmetry}) \tag{7.12}$$

If ζ_{min} is not met by the load alone, add a damping network (R_d–C_d branch with $n = 3 \ldots 5$ is the usual choice; R_p–L_p across L if the DC-side capacitance must stay small, see [5]) or provide active damping in the current controller. The controller bandwidth of a line-commutated bridge is limited by the firing delay ($T_d \approx 1/(12f) \ldots 1/(6f)$), so f_c is typically 10–30 Hz at 50 Hz; the window $f_c \ll f_0 \ll 6f$ is narrow and must be checked explicitly.

7.5 Component ratings, precharge

Inductor: $I_{L,dc} = I_d, \qquad I_{L,rms}^2 = I_d^2 + \frac{1}{2}\Delta\hat{I}_L^2, \qquad I_{sat} > \max(i_p,\ I_{ol}), \qquad W_L = \frac{1}{2} L I_d^2$ (7.13)

Capacitor: $V_C \geq k_C\, V_{d0}(1 + \Delta V_+),\ k_C = 1.1 \ldots 1.2 \ (\alpha \to 0 \text{ control fault})$ (7.14)

$$I_{C,rms} \approx \frac{\Delta\hat{I}_L}{\sqrt{2}} \ (\text{dominant harmonic}), \qquad P_C = I_{C,rms}^2\, ESR, \qquad \Delta v_{C,pp} = 2A_6|H(\omega_6)| \tag{7.15}$$

Load step: $\Delta\hat{v}_C \approx \Delta I\sqrt{\frac{L}{C}}$ (load rejection, undamped; reduced by damping and control) (7.16)

Hard start: $\hat{I}_L \approx V_{d0}\sqrt{\frac{C}{L}} + I_d$ ($\alpha = 0$; check I_{TSM}, fuse I^2t, I_{sat}; else soft-start from $\alpha \approx 90°$) (7.17)

8 Grid-side quantities

$$I_2 = 0.816\, I_d, \qquad I_1 = \frac{\sqrt{6}}{\pi} I_d = 0.780\, I_d, \qquad I_h = \frac{I_1}{h},\ h = 6k \pm 1 \text{ (ideal: } \mu = 0\text{, smooth } I_d\text{)} \tag{8.1}$$

$$THD_i = \sqrt{\frac{I_2^2}{I_1^2} - 1} = 0.311, \qquad S_1 = \sqrt{3}\, V_2 I_1, \qquad Q_1 = S_1 \sin\varphi_1 \text{ (compensation / AC filter design)} \tag{8.2}$$

Overlap reduces the high-order I_h; DC ripple current increases the 5th and 7th. Grid-side power $P_{ac} = V_d I_d + P_{loss}$.

9 Rules of thumb

Quantity	Typical value	Note
α_{min}	5–10°	firing reliability, control headroom at $V_{1,min}$
γ_{min} (inverter)	15–20°	$\geq \omega t_q$ + margin
d_x, d_r, D_t	$u_k/2$, $1.05\, u_r$, 2–3 V	first-pass drops
S_T	$1.05\, P_{d0}$	B6, both windings
$k_v = V_{RRM}/\hat{V}_{2,max}$	2.0–2.5	2.0 only with verified clamping and $\alpha + \mu \ll 90°$
T_j design margin	15–25 K below $T_{j,max}$	plus power-cycling check
Snubber design point	$\alpha + \mu = 90°$	if soft start or current limit reach it
$k_{rep} = V_{RM}/V_{RRM}$	0.5–0.6	repetitive peak at turn-off
ζ_s, R_s	0.5–0.8, $(1 \ldots 1.6)\, Z_0$	$\zeta_{opt}(\chi)$, Table 1
C_s	$2L_k\, (I_{RM}/\chi_{max} V_{com})^2$	energy bound 3–5× larger
$(dv/dt)_{max}$	$R_s \hat{V}_{2,max}/(2L_k)$	$\leq (dv/dt)_{crit}/2$
P_{Rs}	$\leq f\,(1.75\, C_s \hat{V}_2^2 + E_L)$	maximum at $\alpha \approx 90°$
L_σ (snubber loop)	tens of nH	at the device, inside the fuse
AC side	RC or MOV for $W_m = i_0 S_T/(2\omega)$	arm snubbers alone insufficient
k_F (fuse derating)	1.2–1.5	temperature, cooling, cyclic duty
k_s (I^2t margin)	0.7–0.8	fuse I^2t_{tot} at actual U_2
$\hat{U}_{arc}$	$\leq V_{RSM}$	conservative
f_0	$\leq 2f$ (i.e. $6f/3$), away from f, $2f$	attenuation ≥ 8 at $6f$
ζ_{min}	0.3–0.5	at minimum load
L	$(1.2 \ldots 1.5)\, L_{min}$	continuous conduction at $I_{d,min}$
V_C	$(1.1 \ldots 1.2)\, V_{d0}(1 + \Delta V_+)$	control fault at $\alpha \to 0$
THD_i	31 %	ideal six-pulse; $I_h \approx I_1/h$
Start	soft-start from $\alpha \approx 90°$	or precharge circuit

10 Worked example

Specification: $V_d = 440$ V, $I_d = 1000$ A, $I_{d,min} = 100$ A, overload 1.5× for 60 s, grid 400 V ± 10 %, 50 Hz, transformer $u_k = 6$ %, $u_r = 1$ %, $i_0 = 1.5$ % (assumed), $\alpha_{min} = 10°$, $r_{pp} = 3$ %, soft start from $\alpha \approx 90°$. Representative thyristor data (not a specific part): $V_{T0} = 0.85$ V, $r_T = 0.35$ mΩ, $R_{thJC} = 0.030$ K/W, $R_{thCH} = 0.010$ K/W, $I_{TSM} = 15$ kA (10 ms) ⇒ $I^2t_T = 1.13 \times 10^6$ A²s; $Q_{rr} = 1.0$ mC at 10 A/µs and $T_{j,max}$, $S = 0.3$, $(dv/dt)_{crit} = 500$ V/µs, $(di/dt)_{crit} = 150$ A/µs. All figures are reproduced by `b6c_example_calc.py`; `b6c_snubber_sim.py` simulates the bridge with its snubbers.

Step	Equation	Result
S2–S3	(2.6), (2.1), (2.7), (2.8), iterated	$V_{d0} = 523$ V (first guess 447 V, 5 iterations to 1 mV); $V_2 = 387$ V, $n_{tr} = 1.033$; $I_2 = 816$ A, $S_T = 548$ kVA; $X_k = 16.4$ mΩ, $L_k = 52$ µH, $R_k = 2.7$ mΩ
Drops	(2.3)–(2.5)	$D_x = 15.7$ V ($d_x = 3.0$ %), $D_r = 5.5$ V ($d_r = 1.05$ %), $D_t = 2.4$ V
S4	(3.2), (3.1), (3.3) at $V_1 = 360/400/440$ V	$\alpha = 10.0/27.6/36.3°$; $\mu = 13.3/6.7/5.0°$; $\cos\varphi_1 = 0.951/0.856/0.778$; $Q_1 = 142/265/355$ kvar (taps or compensation advisable)
S5	(4.1)	$\hat{V}_{2,max} = 602$ V; $V_{RRM} \geq 1205$ ($k_v = 2$) … 1506 V ($k_v = 2.5$) ⇒ 1600 V class ($k_v = 2.66$; 1800 V if $\hat{U}_{arc}$ or the snubber require)

Step	Equation	Result
S6	(4.3), (4.4), (4.5)–(4.7)	$I_{T(\mathrm{AV})} = 333\,\mathrm{A}$, $I_{T(\mathrm{RMS})} = 577\,\mathrm{A}$, $P_T = 400\,\mathrm{W}$; overload 500/866 A, $P_{ol} = 688\,\mathrm{W}$; $T_j = 86\,^\circ\mathrm{C}$ at $T_{hs} = 70\,^\circ\mathrm{C}$, 98 °C at overload ($T_{hs}$ constant); $R_{thHA} \le 0.135\,\mathrm{K/W}$ per device for $T_a = 40\,^\circ\mathrm{C}$, $T_j \le 110\,^\circ\mathrm{C}$
S7	(3.6)	$(di/dt)_{on} = 0.8 \ldots 3.4\,\mathrm{A/\mu s}$ ($\alpha = 10 \ldots 36^\circ$), plus the snubber discharge of S8 ($I_{dis} \le 73\,\mathrm{A}$)
S8	(5.1)–(5.3)	Design point $\alpha + \mu = 90^\circ$ (soft start, current limit) at $I_d = 1000\,\mathrm{A}$: $V_{com} = 602\,\mathrm{V}$, $(di/dt)_{off} = 5.8\,\mathrm{A/\mu s}$, $Q_{rr} = 0.76\,\mathrm{mC}$, $I_{RM} = 82\,\mathrm{A}$, $E_L = 0.35\,\mathrm{J}$
	(5.6)–(5.8), Table 1	$V_{RM,adm} = 0.55 \times 1600 = 880\,\mathrm{V}$, $\rho = 1.46 \Rightarrow \chi_{max} = 0.94$, $C_s \ge 2.2\,\mu\mathrm{F}$ (energy bound 9.1 µF) $\Rightarrow C_s = 2.7\,\mu\mathrm{F}$; $Z_0 = 6.2\,\Omega$, $\chi = 0.85$, $\zeta_{opt} = 0.72$ $\Rightarrow R_s = 8.2\,\Omega$ ($\zeta_s = 0.66$, $f_{0s} = 9.5\,\mathrm{kHz}$); $V_{RM} = 847\,\mathrm{V} = 0.53\,V_{RRM}$
	(5.9), (5.10)	$(dv/dt)_{max} = 47\,\mathrm{V/\mu s}$ ($R_s \le 43\,\Omega$ with $k_{dv} = 2$); $I_{dis} = 73\,\mathrm{A}$, $E_{dis} = 0.49\,\mathrm{J}$; at the S4 points ($\alpha + \mu = 23/34/41^\circ$): $V_{RM} = 314/465/582\,\mathrm{V}$
	(5.11)–(5.13)	$P_{Rs} = 9/25/43\,\mathrm{W}$ at the S4 points, 103 W at $\alpha = 90^\circ$ (six snubbers: 0.26 kW in steady state); $I_{Cs,rms} = 2.3\,\mathrm{A}$ (3.6 A at 90°), $\hat{\imath}_C = 82\,\mathrm{A}$, $E_{pulse} = 0.84\,\mathrm{J} \Rightarrow C_s$: 2.7 µF PP film, ≥ 1.2 kV DC, snubber-rated; R_s: 8.2 Ω, 100 W class, non-inductive
	Alternative	1800 V class: $V_{RM,adm} = 990\,\mathrm{V}$, $\rho = 1.64 \Rightarrow C_s = 1.5\,\mu\mathrm{F}$, $R_s = 10\,\Omega$, $V_{RM} = 967\,\mathrm{V}$, $P_{Rs} = 28\,\mathrm{W}$ (65 W at 90°)
	(5.14), (5.15)	$W_m = 13\,\mathrm{J}$; arm snubbers alone ($C_Y = 2C_s = 5.4\,\mu\mathrm{F}$): up to 3.2 kV ⇒ AC-side RC with $C_Y \ge 49\,\mu\mathrm{F}$ per phase for $\hat{V}_{max} = 0.75\,V_{RRM} = 1.2\,\mathrm{kV}$ ($Q_C = 2.3\,\mathrm{kvar}$), or MOVs ≥ 13 J per operation with margin
S9	(6.5)–(6.8) at $V_{1,max}$	$Z_k = 16.7\,\mathrm{m\Omega}$, $X/R = 6$; $I''_{k3} = 14.8\,\mathrm{kA}$, $I''_{k2} = 12.8\,\mathrm{kA}$; $\kappa = 1.61$, $i_p = 29\,\mathrm{kA}$; device-fault $di/dt = 5.8\,\mathrm{A/\mu s}$; DC-side short $di/dt \approx 0.55\,\mathrm{A/\mu s}$ with $L = 1\,\mathrm{mH}$
S10	(6.1)–(6.3)	$I_{Fn} \ge 1.3 \times 577 = 751\,\mathrm{A} \Rightarrow 800\,\mathrm{A}$; $U_{Fn} \ge 426\,\mathrm{V} \Rightarrow 500$ or 690 V class; $I_{cn} \ge 15\,\mathrm{kA}$
S11	(6.10)–(6.13)	Requirement: $I^2t_{tot,F}(426\,\mathrm{V}, 12.8\,\mathrm{kA}) \le 0.8 \times 1.13 \times 10^6 = 0.90 \times 10^6\,\mathrm{A^2s}$, $\hat{\imath}_c \le 15\,\mathrm{kA}$, $\hat{U}_{arc} \le V_{RSM}$, $t_m(866\,\mathrm{A}) \ge 3$–5 min – verify on the fuse curves
S12	(7.1), (7.3), (7.5), (7.7)	$\alpha_{max} = 36.3^\circ$, $V_{d0} = 575\,\mathrm{V}$: $A_6 = 120\,\mathrm{V}$, $A_{12} = 57\,\mathrm{V}$; $\Delta\hat{v}_{target} = 6.6\,\mathrm{V} \Rightarrow$ $LC \ge 5.4 \times 10^{-6}\,\mathrm{s^2}$ ($f_0 \le 69\,\mathrm{Hz}$); $L_{min} = 0.64\,\mathrm{mH} \Rightarrow L = 1.0\,\mathrm{mH}$, $C = 5.6\,\mathrm{mF}$, $f_0 = 67\,\mathrm{Hz}$
Check	(7.2), (7.9)	$\lvert H(6f)\rvert = 0.052$: ripple 12.4 V pp = 2.8 %; 12th adds 0.7 V; $\zeta = 0.48$ at full load, 0.05 at 10 % load ⇒ damping network required; $\lvert H(f)\rvert = 1.2$ (2.2 at 10 %), $\lvert H(2f)\rvert = 0.53$ (0.82): components at f, $2f$ pass – limit unbalance/asymmetry or move f_0 (e.g. ≤ 35 Hz with $C \approx 20\,\mathrm{mF}$)
S13	(7.13)–(7.17)	$\Delta\hat{I}_L = 67\,\mathrm{A}$, $I_{C,rms} \approx 47\,\mathrm{A}$; $V_C \ge 1.1 \times 575 = 633\,\mathrm{V} \Rightarrow$ 700–800 V class; hard start $\hat{I}_L \approx 1360 + 1000 = 2360\,\mathrm{A} \Rightarrow$ soft-start; undamped load rejection up to 423 V ⇒ damping + control
S14	(8.1)–(8.2)	$I_1 = 780\,\mathrm{A}$, $I_5 = 156\,\mathrm{A}$, $I_7 = 111\,\mathrm{A}$, $I_{11} = 71\,\mathrm{A}$, $THD_i = 31\,\%$

The example shows the typical iterations. A 3 % ripple target at $\alpha_{max} \approx 36^\circ$ forces $f_0 \approx 67\,\mathrm{Hz}$, between f and $2f$; the alternatives are a lower α_{max} (primary taps, so that the filter is sized at a smaller A_6), a larger C with $f_0 < 40\,\mathrm{Hz}$, or an accepted higher ripple with $f_0 > 2.5f$. The snubber is sized by the soft start, which takes the bridge through $\alpha + \mu = 90^\circ$: with the 1600 V class the budget $\rho = 1.46$ asks for 2.7 µF per arm, while the 1800 V class roughly halves capacitance and losses, so the voltage class is best chosen together with the snubber. The arm snubbers cannot absorb the magnetizing energy of the transformer; the AC side needs its own RC network or MOVs. The fuse figures are requirements to be checked on the selected fuse curves, not results; the snubber figures rest on the ideal snap-off and on a scaled Q_{rr}, and are to be confirmed by simulation with a reverse-recovery device model and by measuring v_T at commissioning.

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